Zak-OTFS Channel Estimation Project Report

From Original Paper Reproduction to Structure-Aware Distilled Compression

Indian Institute of Technology Jodhpur
Department Project Report

WS01 - Emerging Modulation Techniques Towards 6G Networks 2025

Zak-OTFS with Superimposed Spread Pilot: CNN-Aided Channel Estimation

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***Abstract*—Accurate estimation of the effective channel is crucial for reliable data detection in Zak-OTFS systems. Channel estimation using exclusive or embedded pilot frames with point pilot incurs throughput loss. In addition, frames with point pilot exhibit high peak-to-average power ratios. Superimposed spread pilot-based channel estimation can alleviate these issues. Conventional channel estimation using superimposed spread pilot involves computing the cross-ambiguity function and estimating the effective channel taps from a lattice that is twisted with respect to the period lattice on which the data signal is supported. However, this approach is affected by delay-Doppler (DD) domain aliasing in estimating the effective channel, and the estimation is further affected due to interference from the data signal and receiver noise. This compromises the estimation quality, which affects the detection performance. In this work, we propose a learning-based framework to enhance the estimation performance. Specifically, we propose a novel approach that treats the cross-ambiguity read-off as a two-dimensional (2D) image and uses a convolutional neural network (CNN) designed to enhance this image by jointly suppressing the DD aliasing and data interference, and denoising. Our simulation results with Vehicular-A channel model with fractional delays and Dopplers demonstrate that the proposed CNN-aided solution outperforms the conventional estimation method in terms of normalized mean squared error and bit error rate (BER), and achieves a BER performance close to that with perfect channel knowledge.**

***Index Terms*—Zak-OTFS, superimposed spread pilot, channel estimation, cross-ambiguity, convolutional neural network.**

I. Introduction

Orthogonal time frequency space (OTFS) modulation, a modulation in the delay-Doppler (DD) domain, is well-suited for doubly-selective channels which are typical in 6G and beyond [1],[2]. Recently, a new version of OTFS using Zak transform, called Zak-OTFS, has been introduced in [3],[4]. In Zak-OTFS, the information-bearing DD domain signal is directly transformed into the time domain using the inverse Zak transform. Zak-OTFS has the advantages of having a formal mathematical framework using Zak theory to describe the Zak-OTFS waveform and better performance compared to the earlier multicarrier version of OTFS [3], [4], [5]. Motivated by these advantages, research on Zak-OTFS is gaining momentum [6]-[13].

Accurate channel estimation is crucial for reliable data detection in Zak-OTFS receivers. Three types of pilot frames are common for channel estimation in the Zak-OTFS literature: i) exclusive pilot frame consisting of a point pilot and no data symbols [4], [11], [13], $ii)$ embedded pilot frame consisting of a point pilot and data symbols with guard space in between [10]-[12], and $iii)$ superimposed spread pilot frame consisting of data symbols and a spread pilot superimposed on each other [6]. Exclusive and embedded pilot frames have the drawbacks of throughput loss due to unused guard space and high peak-to-average power ratio (PAPR) due to point pilot. In the superimposed spread pilot scheme [6], a spread pilot is obtained by applying a discrete DD spreading filter to a point pilot, and this spread pilot signal is superimposed on the data signal. Multiplexing of data and spread pilot in the same frame eliminates the loss of throughput, and the spreading operation distributes the pilot energy over the entire frame, which reduces the PAPR. However, this scheme faces the challenge of carrying out channel estimation in the presence of data interference and noise. In this paper, we are concerned with enhancing the channel estimation performance of Zak-OTFS with superimposed spread pilot using a novel learning-based approach, which has not been reported before.

The conventional method of estimating the channel with superimposed spread pilot in Zak-OTFS exploits the nature of the self-ambiguity function of the spread pilot. The self-ambiguity function of the spread pilot is supported on a twisted lattice Λ_t, which is rotated with respect to the period lattice Λ_p on which the data signal is supported [6]. At the receiver, the channel estimate is obtained by computing the cross-ambiguity between the received signal and the spread pilot and simply reading off the cross-ambiguity output within the support set. The cross-ambiguity output has the effective channel supported on the self-ambiguity lattice Λ_p. The read-off samples consist of (i) the true effective channel taps, (ii) the DD aliasing components, (iii) interference from data signal, and (iv) noise. When the terms (ii)-(iv) are significant (which depends on the characteristics of pulse shaping filter and the physical channel response), the estimation quality is compromised. Recognising this phenomenon in the conventional method, we propose a novel enhancement technique.

The novelty of the proposed technique lies in viewing the cross-ambiguity read-off as a 2D image and using a convolutional neural network (CNN) designed to enhance this image by jointly suppressing the DD aliasing (term (ii)) and data interference (term (iii)), and denoising (term (iv)). Our simulation results with Vehicular-A channel model [14] with fractional delay and Doppler spreads demonstrate that the proposed CNN-aided solution outperforms the conventional estimation method in terms of normalized mean squared error (NMSE) and bit error rate (BER), and achieves a BER performance close to that with perfect channel knowledge.

II. Zak-OTFS System Model with Spread-Pilot

Figure 1 shows the Zak-OTFS transceiver with superimposed spread pilot. The basic information carrier in Zak-OTFS

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First page of the original PIMRC 2025 paper that motivated this project.

Team Members

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Report Scope. This report explains the original Zak-OTFS paper, our independent baseline reimplementation, the structural novelty we added, and the final comparison between the original method, our baseline, and our distilled Lite models.

Deliverables Covered

- Original-paper explanation in clear engineering language
- Baseline reimplementation and comparison with the published method
- Novelty: physics-induced support compressibility and distilled Lite-M model
- Quantitative comparison using the validated results available in our pipeline
- Team contribution split

Prepared from the reproduced Zak-OTFS baseline, the distilled-student extension, and the final structural-compressibility analysis.

1. Original Paper: What It Does and Why It Matters

The original paper, "Zak-OTFS with Superimposed Spread Pilot CNN-Aided Channel Estimation," studies channel estimation for Zak-transform-based OTFS in a highly time-varying wireless setting. Its key engineering idea is that a spread pilot can be superimposed on the delay-Doppler (DD) grid without the guard-band overhead used in earlier embedded-pilot OTFS schemes. The receiver reads out a compact twisted-support channel image from the cross-ambiguity response and then refines that image using a convolutional neural network (CNN).

The paper matters for two reasons. First, it targets a physically meaningful high-mobility regime where accurate DD-domain channel estimation is difficult. Second, it proposes a CNN-aided improvement over the conventional read-off estimate without changing the overall Zak-OTFS detector structure. In other words, the learned part sits at a very important junction: if the support image is improved, both pilot cancellation and downstream data detection improve.

At the signal-model level, the original method is built around the self-ambiguity structure of Zak-OTFS with Gaussian-shaped pulses. Instead of allocating a separate guard region for pilots, the transmitted frame carries both information symbols and a spread pilot together. The receiver then computes a pilot-based read-off over a carefully defined twisted-support crop. This crop is much smaller than the full DD grid, which is why the paper can reinterpret channel estimation as a compact image-restoration problem rather than a full-grid reconstruction problem.

This is also why the original paper is interesting from a machine-learning perspective. The CNN does not estimate an arbitrary wireless channel from scratch; it denoises a structured, low-resolution complex support image that already contains physically meaningful geometry. The published work therefore sits at the intersection of communication-theory structure and learned image-style refinement, which made it a good candidate for both faithful reproduction and later compression.

Original paper in one line. The published method treats the twisted-support DD channel estimate as a small complex image and uses a CNN to denoise it before LMMSE detection.

At a high level, the original pipeline is:

1. Superimpose data and a spread pilot in the Zak-OTFS DD grid.
2. Pass the signal through a sparse Vehicular-A channel with fractional delay and Doppler shifts.
3. Read off a compact support-domain channel image from the pilot response.
4. Use a CNN to refine the noisy support image.
5. Re-embed the refined estimate into the DD grid and perform pilot cancellation plus LMMSE data detection.

The original paper reports that its CNN-aided estimate approaches the perfect-CSI reference under its published evaluation setting. However, no official code accompanied the paper, so any later implementation effort must be framed as a careful reimplementation rather than a byte-for-byte reproduction of the authors' internal code.

From a project standpoint, this missing-code issue matters a lot. It means we have to separate three things very clearly:

- what the original paper *states* explicitly,
- what we can reproduce directly from those stated equations and settings,
- and what must be implemented through documented engineering assumptions.

That distinction is the reason our baseline and novelty sections are written conservatively. We are extending a paper-defined method, not reverse-engineering a hidden official repository.

2. Our Baseline: What We Rebuilt, Why We Rebuilt It, and How It Compares

2.1 What We Rebuilt

Because the original paper did not provide an official repository, we rebuilt the full workflow ourselves:

- Zak-OTFS DD-grid construction and spread-pilot generation
- Vehicular-A channel simulation with fractional delay-Doppler shifts
- Cross-ambiguity-based support read-off

- The published CNN architecture for support-image refinement
- Pilot cancellation and LMMSE detection

The most important baseline choice was to preserve the *published estimator design*: the same four-layer CNN, large-kernel image-processing view of the support crop, and the same overall inference path from support estimation to detection.

2.2 Why the Baseline Was Necessary

The baseline is not just background code. It is the foundation that makes the later novelty defensible. Without it, any compact model would have no trustworthy teacher, no fair evaluation path, and no way to separate true novelty from implementation noise. The baseline also forced us to document where the original paper was underspecified, especially around the effective-channel construction and matched-filter discretization.

2.3 How Our Baseline Compares with the Original Paper

Table 1: Original paper vs. our baseline reimplementation.

Aspect	Original paper	Our baseline
Code availability	No official public repository	Full independent implementation
Estimator	CNN-aided support-image denoiser	Same estimator structure reproduced
CNN size	Large four-layer CNN	Same verified architecture: 245,473 params
Physics path	Described in paper	Fully implemented end-to-end
Detector	Pilot cancellation + LMMSE	Same detection path retained
Numerical comparability	Published results only	Internally consistent within our implementation, but not claimed as exact author-code parity

Our baseline is faithful in *structure* rather than guaranteed identical in every hidden implementation detail. That distinction matters. It means we can compare teacher and student models honestly inside our own pipeline, while avoiding the unsafe claim that we exactly reproduced the unpublished author code.

3. Novelty We Added: Structural Compressibility + Distilled Lite Models

3.1 The Core Insight

Our novelty is **not** merely "we compressed a CNN." The real novelty is that we found a physical reason why compression should work well for this task.

The support-domain Zak-OTFS channel image is not an arbitrary image. It comes from a sparse wireless channel with only a few physical paths. In our default setting, the channel has $P = 6$ paths. Each path contributes a structured delay-Doppler pattern to the support crop, and the full image is formed by adding only a small number of such patterns together. That already suggests that the target image should be much simpler than a generic image denoising problem.

There is also a deterministic chirp embedded in the support image. When we remove this known chirp by dechirping, the true support map becomes much simpler to analyze. Under the fast effective-channel model used in our implementation, the dechirped support image can be written as a sum of path-wise separable delay and Doppler atoms. In matrix language, it becomes a sum of outer products. Since there are only P physical paths, the rank is bounded by P.

In simpler words, the true channel image is hiding inside a small, structured, low-dimensional family. The CNN is therefore not learning an unrestricted image-restoration task. It is mostly learning how to push a noisy observed support image back toward this physically constrained family of valid support images. Once we understand that, strong compression becomes much easier to justify: a smaller student does not need to reproduce every behavior of a large generic CNN, it only needs to learn the denoising map on a structured manifold.

Main novelty statement. The distilled student works well because the Zak-OTFS support image is not a generic image-denoising problem. After dechirping, the target lies on a low-dimensional physics-induced manifold.

This changes the interpretation of the whole project in a very direct way:

- The teacher CNN is strong, but it is solving a *structured* denoising problem rather than a free-form image task.
- A compact student can imitate that behavior because the space of meaningful target images is already highly constrained.
- Distillation is therefore justified by signal structure, not just by trial-and-error model shrinking.

The picture below summarizes the novelty in the simplest possible way. We start with a noisy support-domain observation that still looks image-like. Once we remove the known support chirp, the hidden structure becomes much easier to see: the true channel no longer behaves like a complicated arbitrary image, but like a small combination of path-driven patterns. That is exactly why a compact student can still work well. The student is learning the map back toward this structured family, not trying to solve an unrestricted image problem.

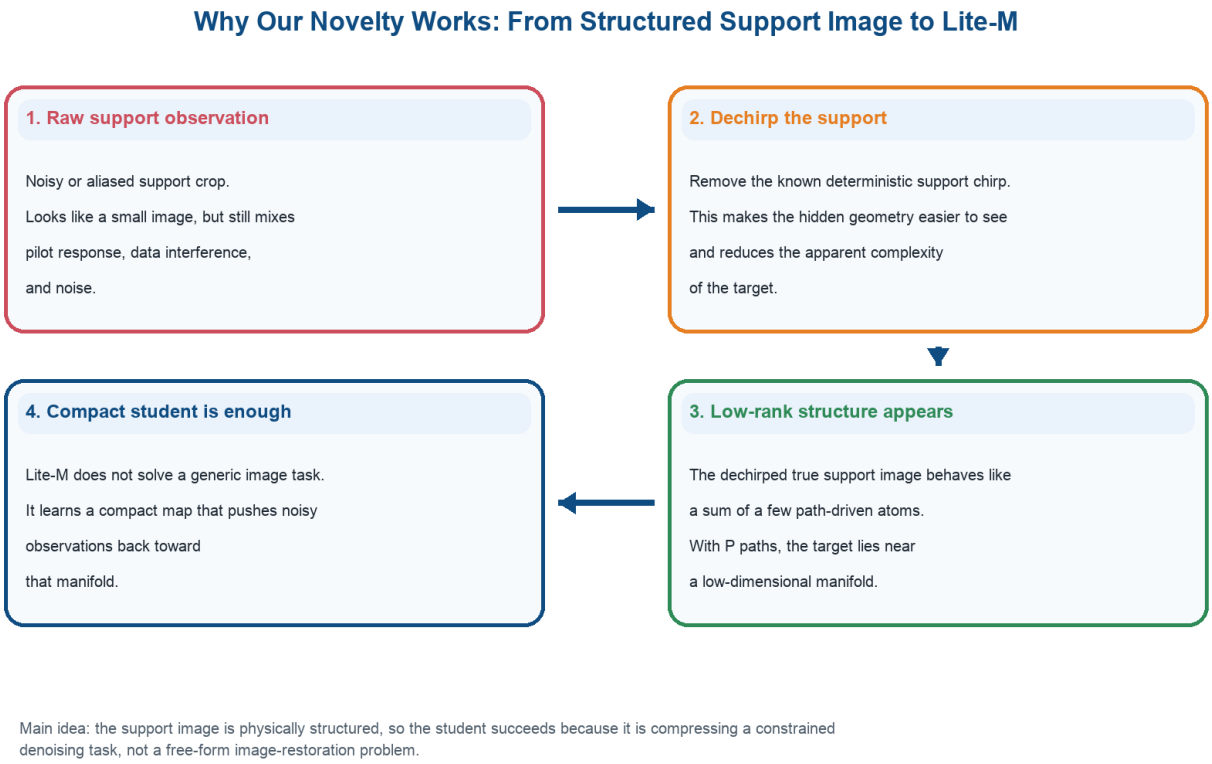

Figure 1: Plain-language view of the novelty. The support image becomes much simpler after dechirping, which explains why a compact student can still recover most of the teacher's behavior.

3.2 Evidence for the Structural Claim

We validated this in three ways:

1. **Exact fast-model structure.** After dechirping, the true support image has negligible singular-value tail energy beyond the path count $P = 6$.
2. **Reference-model support.** The slower reference construction does not admit the same clean symbolic factorization, but numerically it still shows sharply concentrated singular spectra.
3. **Simple non-learned SVD baseline.** A dechirp-plus-truncated-SVD projection already recovers much of the conventional denoising gain, which shows that projection toward a low-rank manifold is doing real work.

These checks matter because they answer the obvious skepticism: what if the student works only because the teacher happens to be redundant in a generic deep-learning sense? Our evidence points to a stronger explanation. The fast model gives an exact algebraic reason for low-rank structure, the slower reference model retains the same concentration trend numerically, and even a simple non-learned projection captures a meaningful part of the gain.

Concrete example from our pipeline. At the representative operating point SNR = 15 dB and PDR = 5 dB, the raw read-off support image has median NMSE 3.31×10^{-1}. If we only dechirp it, keep the top two singular modes, and rechirp it back, the median NMSE drops to 5.70×10^{-2}, which is already very close to the thresholded conventional estimate at 5.84×10^{-2}. This is a direct example showing that a large part of the denoising gain comes from moving the observation back toward a low-rank support manifold.

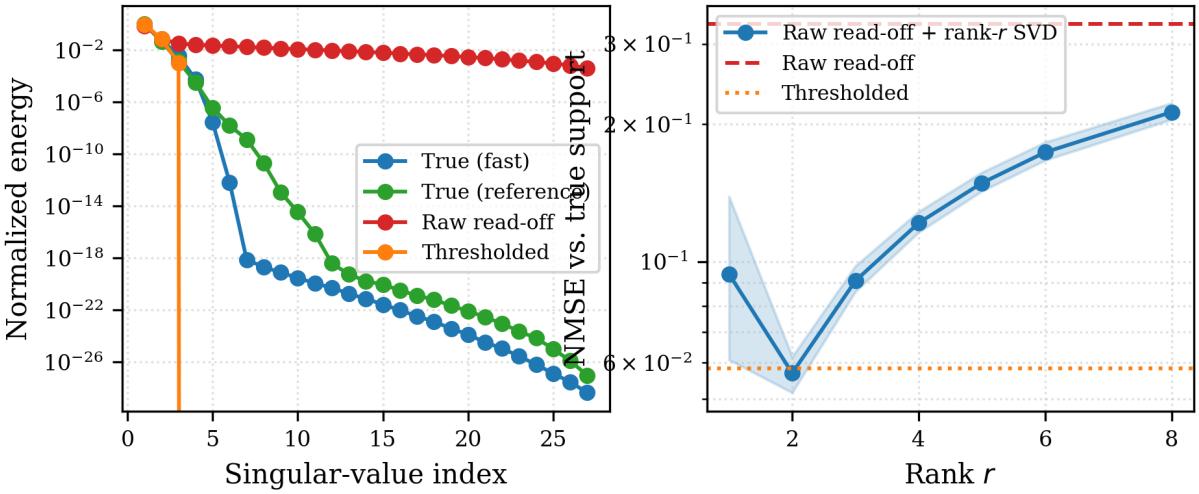

Figure 2: Structural-compressibility evidence. The dechirped true support image is sharply concentrated, while the raw read-off observation has a much heavier singular tail. A simple low-rank projection already recovers much of the denoising gain.

3.3 The Distilled Lite Models

After establishing that the task itself is compressible, we distilled the 245,473-parameter teacher into three compact students:

- **Lite-L:** 40,049 parameters
- **Lite-M:** 22,789 parameters
- **Lite-S:** 6,137 parameters

The student loss combines teacher imitation and ground-truth supervision, so the compact models inherit the teacher's denoising behavior while staying tied to the true support-domain channel. Among the three, **Lite-M emerged as the best overall tradeoff point**.

The role of the three Lite models is also conceptually useful. Lite-L tells us how much compression is possible while staying very close to the teacher. Lite-M identifies the best practical operating point for actual use. Lite-S shows the point where aggressive compression starts to hurt more clearly. This gives the novelty a complete shape: we are not presenting only one smaller model, but an explainable compression frontier grounded in the structure of the Zak-OTFS support image.

So the full novelty can be stated simply as follows: we first explain why the channel-estimation target is compressible, then we use distillation to build compact models that exploit that compressibility in practice, and finally we show that Lite-M gives the best balance between size, speed, and performance.

4. Results and Clear Comparisons

4.1 Original vs. Baseline vs. Our Novelty

The comparison must be stated carefully:

- **Original paper:** introduces the CNN-aided Zak-OTFS estimator and reports strong gains versus the conventional estimator.
- **Our baseline:** independently reconstructs that pipeline with the same overall estimator design and uses it as the teacher model.
- **Our novelty:** explains why compression should work and realizes that idea through a compact distilled Lite-M student.

Table 2: Original paper, our baseline, and our novelty compared.

Aspect	Original paper	Our baseline	Our novelty
Main message	CNN helps DD support estimation	Reproduce the full pipeline faithfully	Explain compressibility and exploit it through distillation
Code	Not publicly available	Fully rebuilt by us	Built on the verified baseline
Explainability	Limited structural explanation	Strong engineering grounding	Physics-induced low-dimensional explanation
Compact model	No	No	Yes, Lite-L / Lite-M / Lite-S
Headline model	CNN estimator	Teacher CNN	Lite-M distilled student

4.2 Quantitative Results

The distilled students retain most of the teacher's practical performance while being far smaller. The figure below combines the four main evaluation curves that matter for this project: NMSE versus PDR, NMSE versus SNR, BER versus PDR, and BER versus SNR.

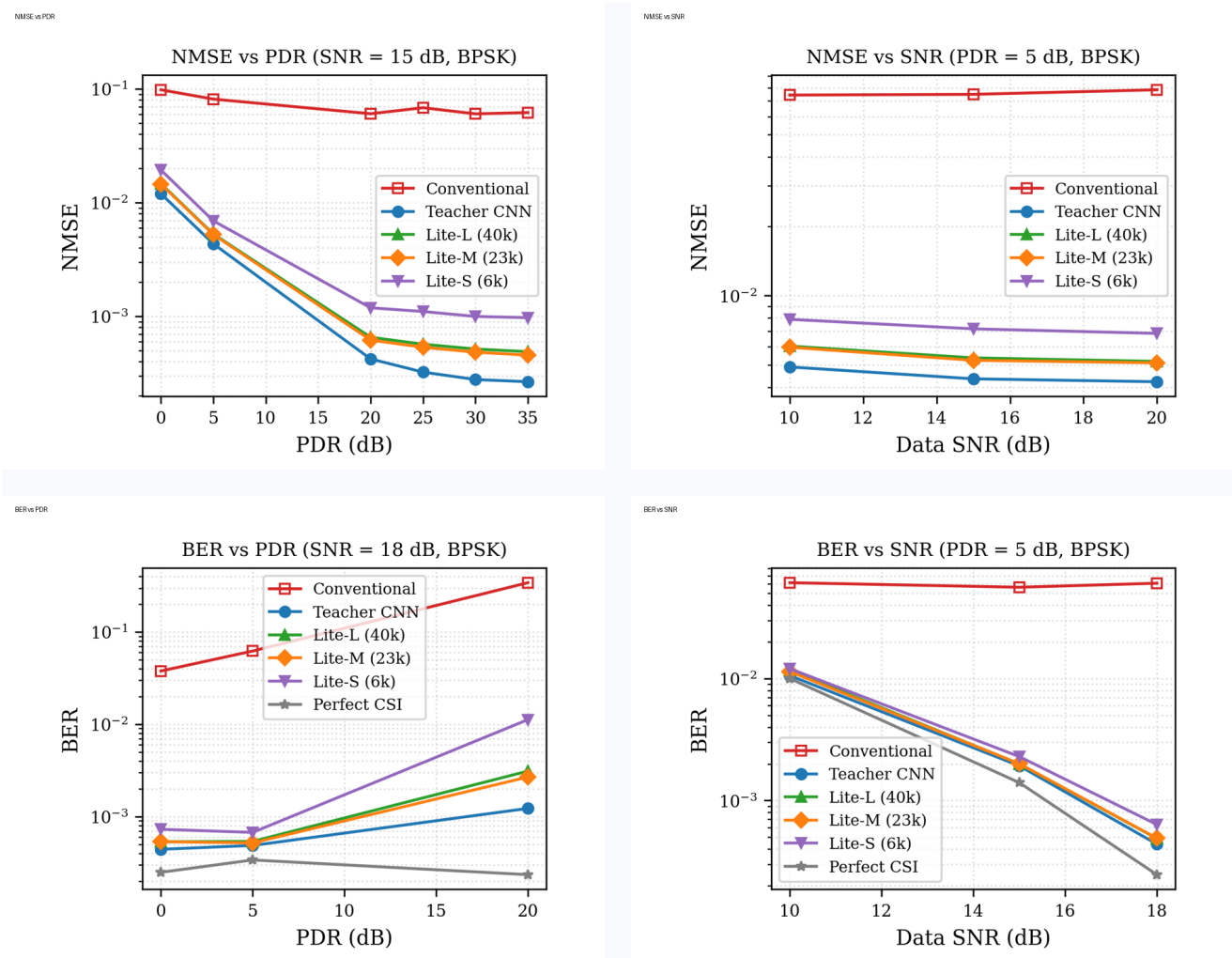

Figure 3: Main performance results for the teacher and distilled students. Lite-M tracks the teacher closely across the retained evaluation suite and is the best overall operating point among the compact models.

The most useful representative operating point is PDR = 5 dB. At that point:

- Conventional estimator NMSE is 8.1410×10^{-2} and BER is 6.0666×10^{-2}.
- Teacher CNN reaches NMSE 4.3520×10^{-3} and BER 4.3899×10^{-4}.
- **Lite-M** reaches NMSE 5.2490×10^{-3} and BER 4.9049×10^{-4} with only 22,789 parameters.

Table 3: Key result summary at the representative PDR = 5 dB operating point.

Method	Params	NMSE	BER
Conventional	—	8.1410e-02	6.0666e-02
Teacher CNN	245,473	4.3520e-03	4.3899e-04
Lite-L	40,049	5.3068e-03	4.8803e-04
Lite-M	**22,789**	**5.2490e-03**	**4.9049e-04**
Lite-S	6,137	6.9299e-03	6.3518e-04

4.3 Why Lite-M Is the Final Model

Lite-L is closest to the teacher, but Lite-M gives the better overall balance. Lite-S is the most aggressive compression point, yet it shows visibly larger degradation under harder conditions. Lite-M is therefore the model that best represents our final contribution:

- strong compression: 10.8× fewer parameters than the teacher,
- practical efficiency gain: about 1.9× measured inference speedup on Apple MPS,
- small accuracy gap at practical operating points.

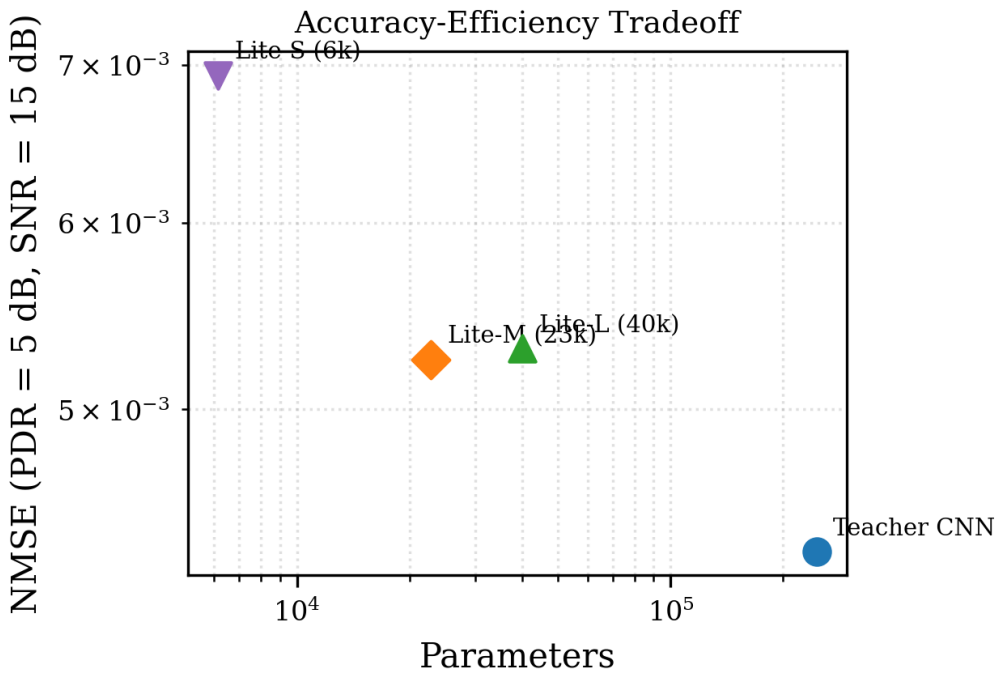

Figure 4: Compression tradeoff. Lite-M is the most balanced point between parameter count and estimation quality.

5. Conclusion and Team Contributions

5.1 Conclusion

This project started from an original Zak-OTFS paper that had no official public code. We first rebuilt the baseline pipeline, including channel simulation, support-domain estimation, CNN refinement, and detector integration. That baseline gave us a trustworthy teacher model and a reproducible evaluation environment.

The main novelty we added is a structural explanation of why compression should work at all. After dechirping, the Zak-OTFS support image is low-dimensional for physical reasons, and this explains why a compact student can imitate a much larger CNN without collapsing in performance. That idea is supported by singular-value analysis, a simple SVD projection baseline, and the final Lite-M results.

The final take-away is clear: **our contribution is not just a smaller CNN.** It is a structure-aware interpretation of Zak-OTFS support estimation, plus a practical distilled model that leverages that structure well enough to retain most of the teacher's value at much lower complexity.

5.2 Contribution Split

To keep the report transparent, the team work is summarized below by responsibility area. Mayank Sharma, Rohit Kumar Mourya, and Jyotiraditya Singh Rathore shared the core technical ownership of the project, while Madhav and Dikshant Jha shared the supporting documentation, validation, and presentation work needed to complete the final deliverables.

Table 4: Team responsibility split.

Roll No.	Name	Primary responsibility areas
B23ES1023	Mayank Sharma	End-to-end baseline integration, final novelty positioning, and report consolidation.
B23ES1029	Rohit Kumar Mourya	Experiment execution, result checking, and original-vs-baseline-vs-novelty comparison analysis.
B23ES1017	Jyotiraditya Singh Rathore	Methodology explanation, structural-compressibility writeup, and final technical narrative support.
B23ES1021	Madhav	Figure curation, documentation support, and formatting cleanup for the final report package.
B23ES1014	Dikshant Jha	Validation passes, artifact organization, and presentation-level polishing of the final deliverables.

Final project message. Original paper: CNN-aided Zak-OTFS estimation. Our baseline: faithful end-to-end reimplementation. Our novelty: physics-induced support compressibility explains why Lite-M can compress the teacher so effectively.

References used in this report come from the reproduced project assets, the original Zak-OTFS paper, and the validated final IEEE draft prepared from the same repository.